

Oksendal 2.6

Let (Ω, F, P) be a probability space and let $A_1, A_2, \dots$ be sets in F such that

$$\sum_{k=1}^{\infty} P(A_k) < \infty$$

Prove the *Borel-Cantelli* lemma:

$$P\left(\bigcap_{m=1}^{\infty} \bigcup_{k=m}^{\infty} A_k\right) = 0$$

i.e. the probability that ω belongs to the infinitely many A_k 's is zero.

Solution

We can start by defining the set we are looking at as E :

$$E = \bigcap_{m=1}^{\infty} \bigcup_{k=m}^{\infty} A_k$$

We can then let $B_m = \bigcup_{k=m}^{\infty} A_k$. This will represent the event that one A_k happens starting from index m . We can notice a nested structure of B_m such that as m increases B_m gets smaller since we are looking at fewer sets of A_k .

$$B_1 = A_1 \cup A_2 \cup A_3 \dots$$

$$B_2 = A_2 \cup A_3 \cup A_4 \dots$$

Since these sets are nesting downwards, the probability of the intersection is the limit of their individual probabilities:

$$P(E) = P\left(\bigcap_{m=1}^{\infty} B_m\right) = \lim_{m \rightarrow \infty} P(B_m)$$

Looking at $P(B_m)$ by the Union Bound (Boole's Inequality), the probability of a union of events is less than or equal to the sum of their individual probabilities:

$$P(B_m) = P\left(\bigcup_{k=m}^{\infty} A_k\right) \leq \sum_{k=m}^{\infty} P(A_k)$$

Since the problem tells us $\sum_{k=1}^{\infty} P(A_k) < \infty$ we know it converges. If a series converges, the tail of the series must go to zero as you start the sum further and further out. So we get the following:

$$\lim_{m \rightarrow \infty} \sum_{k=m}^{\infty} P(A_k) = 0$$

Since $0 \leq P(E) \leq \lim_{m \rightarrow \infty} \sum_{k=m}^{\infty} P(A_k)$, and that limit is 0, the Squeeze Theorem tells us:

$$P(E) = 0$$

Oksendal 2.10

A stochastic process X_t is called stationary if $\{X_t\}$ has the same distribution as $\{X_{t+h}\}$ for any $h > 0$. Prove that Brownian motion B_t has stationary increments, i.e. that the process $\{B_{t+h} - B_t\}_{h \geq 0}$ has the same distribution for all t .

Solution

To prove this we first need to state the three main properties of brownian motion:

1. Starts at zero: $B_0 = 0$
2. Independent increments: For any $s < t$, the increment $B_t - B_s$ is independent of the process up to time s .
3. Normally distributed increments: For any $s < t$, the increment $(B_t - B_s)$ follows a Normal distribution with mean 0 and variance $t - s$. Mathematically: $(B_t - B_s) \sim N(0, t - s)$.

We want to look at the process $Y_h = B_{t+h} - B_t$ for a fixed t and see if its distribution changes when we change t . We need to show that for any h , the distribution of $B_{t+h} - B_t$ depends only on the time interval h , and not on the starting point t .

Using the third property of brownian motion we can define the distribution of the increment over the interval $[t, t + h]$:

- The length of the interval is $(t + h) - t = h$.
- The increment follows: $(B_{t+h} - B_t) \sim N(0, h)$

Looking at the starting increment from $t = 0$ over the same length of time h we get the following:

- The increment is $B_h - B_0$.
- Since $B_0 = 0$, this is just B_h .
- The distribution: $(B_h - B_0) \sim N(0, h)$

Since $(B_{t+h} - B_t)$ and $(B_h - B_0)$ follow the same Normal distribution $N(0, h)$ regardless of the value of t , the distribution depends only on the gap h , and the starting time t does not affect the distribution. This means that brownian motion has stationary increments.